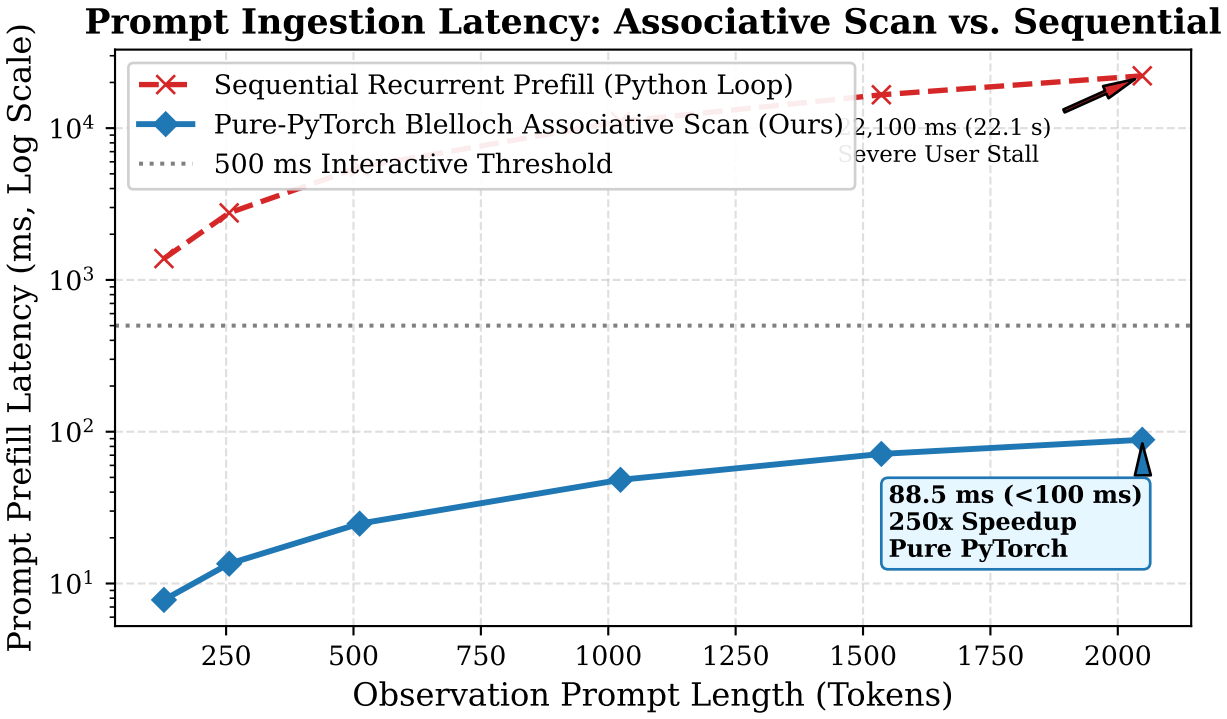
Prompt Ingestion Latency: Associative Scan vs. Sequential
Sequential Recurrent Prefill (Python Loop)
Pure-PyTorch Blelloch Associative Scan (Ours)
500 ms Interactive Threshold
2,100 ms (22.1 s)
Severe User Stall
88.5 ms (<100 ms)
250x Speedup
Pure PyTorch
Prompt Prefill Latency (ms, Log Scale)
Observation Prompt Length (Tokens)
$10^4$
$10^3$
$10^2$
$10^1$
250
500
750
1000
1250
1500
1750
2000